

Coding & Solution

Date	2 October 2026
Team ID	SWTID-2026-8368
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Chippiiiiiii/comic-craft
Programming Language(s)	Python 3.14, HTML, CSS
Framework(s) Used	FastAPI, Jinja2; diffusers (Stable Diffusion v1.5) with PyTorch; google-generativeai; fpdf2
Key Features Implemented	Story form with setting, tone and art-style pickers; Gemini 5-panel outline; Gemini caption and narration; Stable Diffusion panel images; preview page; PDF export; JSON API; /test-image; start-up model warm-up; 24 automated tests
Pending / Incomplete Features	Speech-bubble overlay; character consistency across panels; progress indicator per panel; styled export page; migration to the google-genai SDK
Setup / Run Instructions	python -m venv env; source env/bin/activate; pip install -r requirements.txt; copy .env.example to .env and set GEMINI_API_KEY; uvicorn app.main:app port 8000; open http://127.0.0.1:8000

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes

3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

Measured on Apple M5 (16 GB): about 12.4 s per panel image and 104 s end to end. Gemini free tier allows 5 requests per minute and 20 per day per model, so the outline and story use two different models. google-generativeai is deprecated but was used as specified.